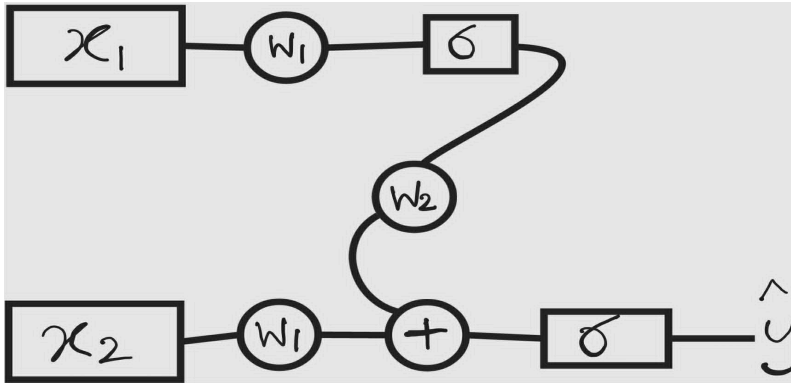


RNN:

Update the weights (w_1 , w_2) of the following RNN after backpropagation



$w_1 = 2$ (input weight)
 $w_2 = 4$ (recurrent weight)
 $x_1 = 0$ (Day 1, older input)
 $x_2 = -1$ (Day 2, recent input)
 $y = 1$ (observed label)
 $\eta = 1$ (learning rate)

Here, sigmoid activation has been used and loss function is cross entropy loss

$$z_1 = 0 \times 2 = 0 \rightarrow h_1 = \sigma(0) = 0.5$$

$$z_2 = (-1 \times 2) + (0.5 \times 4) = -2 + 2 = 0 \rightarrow \hat{y} = h_2 = \sigma(0) = 0.5$$

Loss:

$$L = -\ln(0.5) \approx 0.693$$

Now,

$$\partial L / \partial z_2 = \hat{y} - y = 0.5 - 1 = -0.5$$

$$\partial L / \partial w_2 = \partial L / \partial z_2 \cdot h_1 = -0.5 \times 0.5 = -0.25$$

$$\text{So, } w_{2_new} = 4 - 1 \times (-0.25) = 4.25$$

Now, for calculating $\partial L / \partial w_1$, we need to expand $\partial z_2 / \partial w_1$ across both paths

$$\partial z_2 / \partial w_1 = x_2 + w_2 \cdot \sigma'(z_1) \cdot x_1 = (-1) + 4 \times (0.5 \times 0.5) \times 0 = -1 + 0 = -1$$

$$\text{So, } \partial L / \partial w_1 = \partial L / \partial z_2 \cdot \partial z_2 / \partial w_1 = -0.5 \cdot -1 = 0.5$$

$$\text{So, } w_{1_new} = 2 - 1 \times 0.5 = 1.5$$